IC improve2 item 04: D4 + 2LP regression fix

Item 04: fixing the D_4 + double-large-prime regression, k = 4 IC on E(F_{p^4})

IC improvement round 2 (`../RULES.md`), item 04, 2026-10-15. The protocol, `PROTOCOL.md`, was frozen before any data was collected. Prior work: `../../ic_improve_20261014/01-d4-2lp/`. That item found tuned D_4+2LP / D_4 = 1.025 [1.016, 1.037] on the whole cost, which REGRESSES.

- **Incumbent:** `d4`, the D_4 Y-line arm. Its first LA attempt comes at columns + 1 relations.
- **Candidates:**
  - `d4-ed`: D_4 with the first-dependency stop (from paths 1 and 2);
  - `d4-adapt`: an adaptive α chosen by a cost model (path 3).
- **Reported:** item 01's `d4-2lp` (α = 0.95) and `d4-ed-all` (exploratory).
- **Controls:** a null A/A arm and a positive control.
- **Rho** is a reference column only.

Verdict

The regression is fixed by removing the large primes, not by better large primes. In the D_4 Y-line setting, large primes buy nothing in the relation phase. What the 2LP graph was really doing is finding a linear dependency before columns + 1 relations. If filtering runs after every relation, the full D_4 base gets that saving with no rejected trials.

comparison (test: 27 held-out curves, p = 101…1601)	**phase**	**constant: pooled paired ratio [95 % CI]**	**verdict**	**exponent: Δ slope of $\log_2$ ratio per $\log_2$ #E [95 % CI]**	**verdict**
`d4-ed` / `d4` (candidate A)	**whole**	**0.9792 [0.9725, 0.9855]**	**ADVANCES**	**−0.0008 [−0.0024, +0.0015]**	**MATCHES**
	relation	0.9792 [0.9727, 0.9850]	ADVANCES	−0.0008 [−0.0025, +0.0015]	MATCHES
	linear algebra	0.9693 [0.9552, 0.9827]	ADVANCES	+0.0000 [−0.0036, +0.0052]	MATCHES
`d4-adapt` / `d4` (candidate B)	whole, relation, LA	1.0000 [1.0000, 1.0000]	MATCHES	0	MATCHES
`d4-ed-all` / `d4` (exploratory)	whole	0.9746 [0.9659, 0.9822]	ADVANCES	−0.0010 [−0.0028, +0.0018]	MATCHES
`d4-2lp` / `d4` (item 01's arm, re-measured)	whole	1.0264 [1.0179, 1.0362]	REGRESSES	+0.0047 [+0.0031, +0.0067]	REGRESSES
	linear algebra	0.8201 [0.7979, 0.8456]	ADVANCES	−0.0067 [−0.0129, −0.0008]	ADVANCES
`d4-late110` / `d4` (positive control)	whole	1.1119 [1.0970, 1.1338]	REGRESSES	−0.0057 [−0.0106, −0.0013]	(see note)
`d4-2lp-null` / `d4` (null A/A)	all	1 exactly on 27/27	MATCHES	0	MATCHES

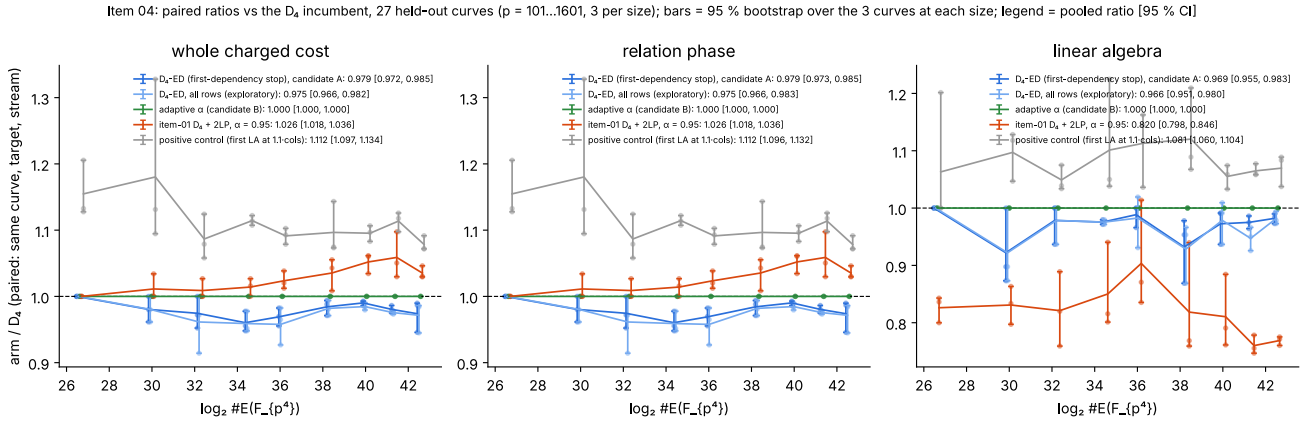
- **Candidate A (`d4-ed`) ADVANCES at the constant level** by 2.1 % [1.5 %, 2.8 %] on the whole cost. Relation and LA phases both advance, so there is no trade.
  - The exponent MATCHES: the saving is a constant fraction of the relations.
  - Per size it is MATCHES at p = 101–269, where the 3-curve CIs touch 1, and ADVANCES at every size from p = 397 to 1601.
  - It never regresses on any curve: per-curve ratio 0.946–1.000.
- **Candidate B (`d4-adapt`)** is **MATCHES at every measured size**, by construction.
  - The frozen model keeps plain D_4 everywhere in the measured range ($\log_2$ #E = 26.6 to 42.6). It is identical to `d4` on 27 of 27 curves.
  - The frozen rule would first leave plain D_4 at **#E ≈ 2^116.5** (α = 0.90, modelled ratio 0.943). This is an extrapolation, not a measurement. Item 01 put its handover at 2^128.9.
- **Item 01's arm** reproduces its regression on the 3 new p = 1601 curves: 1.035 [1.029, 1.046].

Note on the positive control's exponent: `d4-late110` is a known +10 % relation regression. Its ratio is larger at small p, where [1.1·(columns + 1)] rounds up a larger fraction. That gives the negative slope. Its pre-registered test is the constant level: point 1.112, inside [1.04, 1.16], and the CI excludes 1. It **passes**.

Correctness and controls (all passed):

- **Logarithms.** Every logarithm of every arm on every test curve was verified ([d]G = Q and equal to the planted d): 27 × 7 = 189 of 189. On training, all 14 × 6 = 84 were verified. Every decomposition was re-added in the group.
- **Null A/A.** `d4-2lp-null` is identical to `d4` (trials, full relations, LA multiplications, output) on 27 of 27 test curves and 14 of 14 training curves.

- **Reproduction.**
  - `d4` trial counts equal item 01's on all 24 shared test curves and all 14 training curves.
  - α = 0.95 equals item 01's on all 14 training curves.
  - So the new 48-bit bucketed oracle is output-identical to item 01's.
- **ecbench.** The new base `yline-d4-ed` reproduces the direction in the harness (see below). Item 01's committed session still audits clean against the edited code: 36 records, 6 replays reproduced, 0 problems.



paired ratios

Why 2LP cannot fix the relation phase here, and what can

Path 1: large primes, inside or outside the base

(a) Inside the decomposition base B (Thériault; Gaudry–Thomé–Thériault–Diem). In GTTD the large primes are exactly B \ F, the base points left out of a small factor base F with |F| ≪ |B|. Item 01's `d4-2lp-gaudry` arm (|F| = |B|^3/4) is that construction, and it measured 2.36× D₄. Item 01's α = 0.95 arm is the same construction with a larger F. Neither is a "fake" large-prime scheme.

Here is why no choice of F can help the relation phase. Each row is

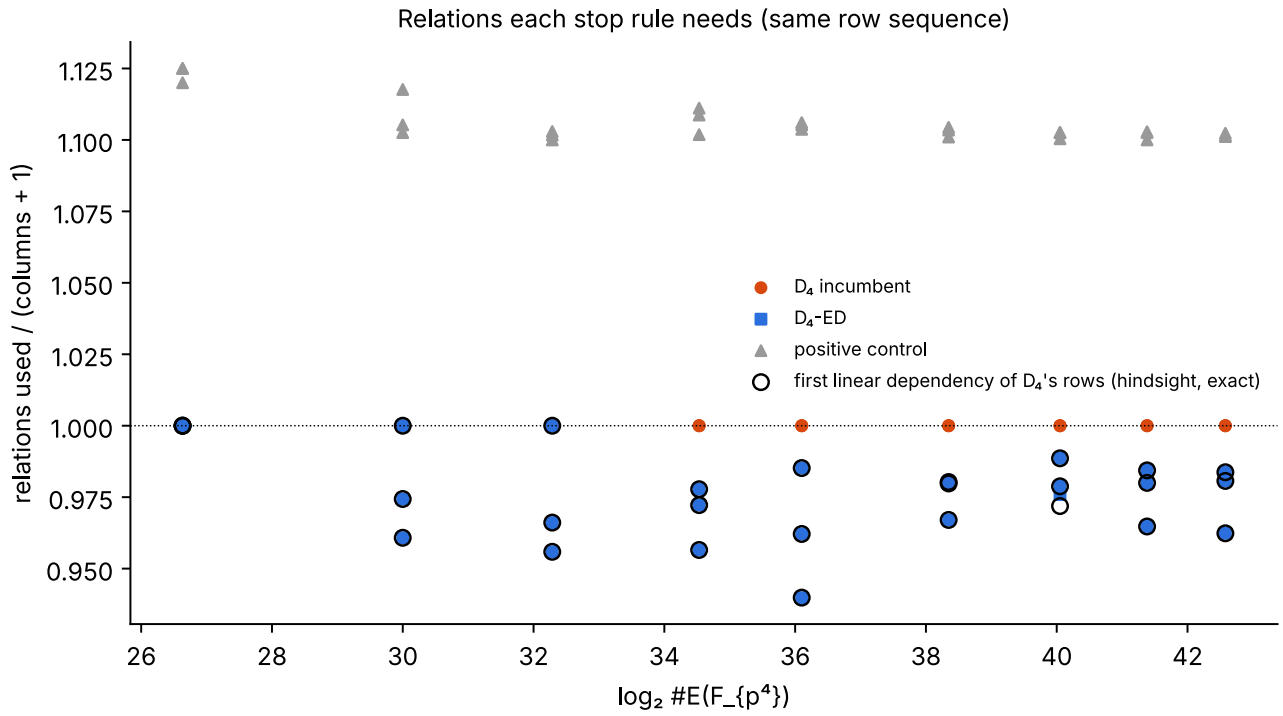
```
Σ_j ±(log P_j) − b·d = a.
```

Every full relation that large-prime elimination (graph cycles, merges) produces is a linear combination of the rows collected so far. So whenever an LP scheme determines d at time m, the collected base rows already have a combination that is zero on every base column and has Σc·b ≠ 0. That is a linear dependency, and full LA would find d from the same rows.

Hence the trials needed by any LP scheme on a shared stream are at least those of the **first-dependency stop (FD)** on the same rows. LP schemes also lose the rows they reject (≥ 3 large primes).

- **Measured.** On every test curve, `d4-ed` and `d4-ed-all` used no more trials than `d4-2lp`. There are 0 violations in 27 curves; `d4-2lp` used 1.000–1.098× D₄'s trials.
- **D₄ orbits.** The orbits {±P, ±(P + T)} are already the columns. Each large-prime "vertex" is an orbit, which halves the vertex set but leaves the argument unchanged.

What the incumbent leaves on the table. D₄ first tries LA at columns + 1 relations. The exact first dependency of D₄'s own row sequence, computed in hindsight by incremental elimination mod r, comes at **0.977 × (columns + 1) relations on average** (test, 27 curves). That is close to the random 4-uniform (XORSAT-type) dependency threshold of about 0.977 rows per column. Columns hit fewer than twice are peeled, which is why the threshold sits below 1. This 2.3 % is the most any stop rule can gain on D₄'s row stream. The square-core stop in `d4-ed` reaches it: it used FD relations on 26 of 27 curves and FD + 1 on one, a mean gap of 0.037 relations.



relations

(b) Large primes outside B (one summand from a set $\Lambda \supsetneq B$, the "much larger outside set"). Let π be the probability that a trial decomposes over B. At p = 1301, $\pi \approx 1/77$.

- Decompositions $R = P_1 + P_2 + P_3 + L$, with $P_i \in B$ and L in a line-sized slice of Λ, occur about 4π per C_4-type solve: 4!/3! orderings. Covering Λ takes $|\Lambda|/|B|$ such slices per target.
- A partial relation is useful only after its L collides. N partials give about $N^2/(2|\Lambda|)$ cycles, so about ncols cycles need $N \approx \sqrt{(2|\Lambda| \cdot \text{ncols})}$ partials.
- **Trials versus D_4:** $\sqrt{(2|\Lambda| \cdot \text{ncols})/(4\pi)}$ against ncols/π, a ratio of $\sqrt{(|\Lambda|/(8 \cdot \text{ncols}))}$.
  - For a 2-dimensional Λ ($|\Lambda| \approx p^2/2$, ncols $\approx$ p/4) this is $\sqrt{(p/4)}$: 18× at p = 1301. It worsens as p^{1/2}, i.e. +1/8 in the exponent in #E.
  - For Λ a second line ($|\Lambda| \approx |B|$), the trial count could halve. But "3 + 1" is then just a larger base with a different decomposition shape. The summands are no longer interchangeable, so the S_4 symmetrisation of the D_4 solver drops to S_3. That multiplies the solution degree by ≥ 4, and C_4 rises by at least that much.
  - Net: ≥ 2× D_4 even if C_4 grew only 4×.
- This is a derivation, not a measurement: C_4 for the 3 + 1 shape was not measured here.

Path 2: stop paying for rejected trials

It cannot be done before the solve. The large-prime count of a decomposition depends on which roots $y_1 \ldots y_4$ the D_4 solve returns. Those come out of the Gröbner basis / FGLM step of the symmetrised system, which is the C_4 cost itself.

- **Resultant-degree filter.** The degree of the eliminant is fixed by the system's shape, not by R, so it carries no information about where the roots lie.
- **Partial solve.** Requiring two summands in F adds $\Pi_{f \in F}(y - y_f)$, of degree |F|, and breaks S_4 to $S_2 \times S_2$. That is more expensive, not cheaper.

The cheap fix is to stop rejecting. `d4-ed` admits every decomposition and finds the dependency with full sparse LA. In the hypergraph sense, every column is a "large prime" and the cycle is the linear dependency.

Path 3: adaptive arm

Fitted on training. Per configuration, the OLS of $\log_2$ relation and $\log_2$ LA GAE on $\log_2$ #E:

- plain D_4: relation slope 0.187, LA slope 0.535;
- $\alpha = 0.90$: relation slope 0.200, LA slope 0.532.

The rule keeps plain D_4 until #E $\approx$ 2^116.5, so at every measured size it is the incumbent (ratio exactly 1). It also flips at 2^{20-2}21.5, below the smallest training size (2^26.3). That is a fit artefact and is not used.

The recommended component is the composition: `d4-ed` as the plain arm, switching to 2LP only past the modelled handover. In the measured range it is `d4-ed`.

Results by size (test split, 3 curves per size)

Paired whole-cost ratios against `d4`: geometric mean, with a 95 % bootstrap over the 3 curves at that size. With 3 curves the per-size CI is coarse; a CI of [1, 1] means all three curves tied.

p	$\log_2$ #E	columns	D_4 trials (mean)	FD / (cols + 1)	d4-ed / D_4	d4-ed-all / D_4	d4-adapt / D_4	d4-2lp / D_4	positive control	D_4 S / rho-neg S (ref.)
101	26.6	23–24	3,315	1.000	1.000 [1.000, 1.000] MATCHES	1.000 [1.000, 1.000]	1.000	1.000 [1.000, 1.000][1]	1.155 [1.128, 1.206]	$1.35 \cdot 10^{8}$
181	30.0	37–50	5,178	0.978	0.980 [0.961, 1.000] MATCHES	0.980 [0.961, 1.000]	1.000	1.011 [1.000, 1.034] MATCHES	1.180 [1.095, 1.328]	$7.25 \cdot 10^{7}$
269	32.3	58–69	7,577	0.974	0.975 [0.952, 1.000] MATCHES	0.961 [0.914, 1.000]	1.000	1.009 [1.000, 1.027] MATCHES	1.087 [1.058, 1.124]	$5.69 \cdot 10^{7}$
397	34.5	89–107	11,294	0.969	0.960 [0.948, 0.978] ADVANCES	0.959 [0.948, 0.978]	1.000	1.014 [1.000, 1.027] MATCHES	1.114 [1.107, 1.122]	$3.80 \cdot 10^{7}$
521	36.1	131–134	10,997	0.962	0.969 [0.955, 0.982] ADVANCES	0.958 [0.927, 0.982]	1.000	1.024 [1.012, 1.039] REGRESSES	1.091 [1.079, 1.103]	$2.18 \cdot 10^{7}$
769	38.3	181–202	19,825	0.976	0.984 [0.971, 0.994] ADVANCES	0.982 [0.971, 0.994]	1.000	1.035 [1.008, 1.055] REGRESSES	1.097 [1.073, 1.143]	$1.98 \cdot 10^{7}$
1033	40.1	248–282	24,831	0.980	0.990 [0.986, 0.993] ADVANCES	0.985 [0.979, 0.990]	1.000	1.052 [1.034, 1.062] REGRESSES	1.095 [1.083, 1.107]	$1.33 \cdot 10^{7}$
1301	41.4	311–349	31,024	0.976	0.980 [0.975, 0.986] ADVANCES	0.975 [0.973, 0.978]	1.000	1.059 [1.030, 1.098] REGRESSES	1.114 [1.098, 1.126]	$1.14 \cdot 10^{7}$
1601	42.6	413–429	34,065	0.976	0.974 [0.946, 0.990] ADVANCES	0.972 [0.945, 0.986]	1.000	1.035 [1.029, 1.046] REGRESSES	1.078 [1.072, 1.092]	$6.31 \cdot 10^{6}$

[1] At p = 101, `d4-2lp` uses the same trials as D_4 and a smaller LA, so the ratio is $1 - 4 \cdot 10^{-10}$. The rule formally calls that ADVANCES. It is an LA saving of 10^{-9} of the cost.

Accounting.

- **Mean relations used / (columns + 1):**

arm	relations / (columns + 1)
d4	1.000
d4-ed	0.977
d4-ed-all	0.977
d4-2lp (full only; pivots extra)	0.795
positive control	1.106

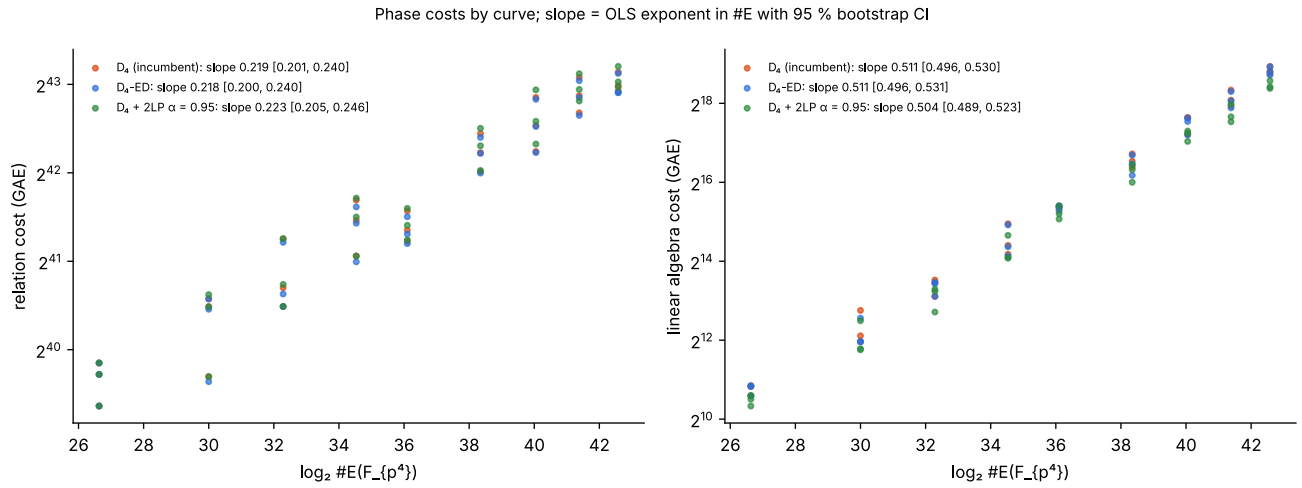
- **`d4-ed-all`'s extra gain** comes from 25 of 5,203 decomposed trials that carried two distinct rows. The trial ratio against D_4 ranges 0.914–1.000, against 0.946–1.000 for `d4-ed`.
- **Every arm solved at its first LA attempt** on every test curve. `d4-ed` runs one Wiedemann solve on a slightly smaller core, hence its LA ratio of 0.969.
- **Filtering bookkeeping.** `d4-ed` calls the square-core filter after every relation: up to $4.0 \cdot 10^{6}$ index operations per curve, uncharged in every arm (PROTOCOL). Charged at 1 F_p mult per index operation, they would be at most $5 \cdot 10^{-9}$ of the whole cost. The whole ratio with them charged is 0.9792 [0.9727, 0.9853], unchanged and still ADVANCES.
- **LA share.** LA is at most $6 \cdot 10^{-8}$ of D_4's whole cost at these sizes.

Fitted exponents in #E

OLS of $\log_2$(cost) on $\log_2$ #E over the 27 test curves, with a 95 % bootstrap over curves.

arm	relation	linear algebra	whole
d4 (incumbent)	0.2186 [0.2012, 0.2396]	0.5107 [0.4956, 0.5304]	0.2186 [0.2024, 0.2406]

arm	relation	linear algebra	whole
d4-ed	0.2178 [0.2001, 0.2402]	0.5108 [0.4963, 0.5309]	0.2178 [0.2001, 0.2395]
d4-ed-all	0.2176 [0.2001, 0.2397]	0.5099 [0.4946, 0.5301]	0.2176 [0.2001, 0.2404]
d4-2lp	0.2233 [0.2049, 0.2457]	0.5041 [0.4888, 0.5226]	0.2233 [0.2061, 0.2445]
d4-adapt	= d4	= d4	= d4



phases

ecbench (in-harness reproduction, 4 sizes)

- **Registration.** Additive edits register the candidate as `ic.k4_gaudry` with `base = yline-d4-ed`:
  - `K4Params.eager_la` in `fp4.rs`: filter after every relation, retry at the next relation;
  - the new base value in `methods.rs`.
- **Hashes.** No parameter was added, so existing specs' method hashes are unchanged.
- **Session.** `ECBS1h08cd110d706f`, spec `ecbench/spec-d4-ed.json`: p = 101, 181, 269, 397 (seed 1) × 3 targets × 2 arms = 24 runs, 24 verified.
- **Audit.** "audit OK: 24 records, 24 verified, 8 replays reproduced, 0 problems". Receipt `ecbench/session-20261015-d4-ed.audit.json`, SHA-256 `49a4ad811469475dbc60198b1de4d093665afbfb30b8d0af665ab9e32a61199f`.
- **Result.** `ecbench compare` gives ic-k4-d4-ed / ic-k4-d4 = **0.9904 [0.977, 0.998]** (cluster bootstrap, 12 workloads).
  - Per curve: 1.000, 0.984, 0.985, 0.980.
  - The harness labels it "diagnostic only (unpriced work)", because the oracle is uncharged.
  - L0 isolation, so these are operation counts only; no wall-time claim is made.
- Full table: `ecbench/compare.txt`.

Deviations from the protocol

1. **Fit output.** The fit's report of where the rule leaves plain D_4 first printed only the first flip, at 2^20.0. That is below the training range and a fit artefact. It was changed before any test data to list every flip; the rule and the model are unchanged.
2. **Exact ties.** The verdict function gained a 10^{-12} tolerance, so that ratios identical to 1 (`d4-adapt`, the null) are called MATCHES as the protocol states. Without it, floating-point rounding of $\exp_2(\text{mean } \log_2)$ called them ADVANCES. This was applied after the test data, and no other verdict changed.
3. **Oracle bucket size.** The first bucket capacity overflowed on a smoke run: D_4 repeats x-values, so bucket counts are clumped. It was enlarged before any recorded run. The overflow is asserted, never silent.
4. **Positive control at small p.** The $\lceil 1.1 \cdot (\text{cols} + 1) \rceil$ rounding makes it larger than 1.10 at small p: 1.155 and 1.180 at p = 101 and 181. The pooled point, 1.112, is inside the registered window.
5. **Machine.** Not the limit for the item.
   - p = 1601 took 11–12 min and 1.95 GB per curve; the oracle build was 6–7 min of that.
   - The host's load average was 8–13 on 4 cores, shared with three other items.
   - p ≈ 2000 would need about 2.9 GB of triple table even at 48 bits (m ≈ 1000 base points), plus the pair table. So 1601 is the largest size that fits the 3 GB cap with this oracle.
   - Training took 6 min of wall time and test 40 min (two lanes).

Reproduce

```
cargo build --release --example ic_improve2_04_d4_2lp_fix --bin ecbench
cd research/ic_improve_20261015/04-d4-2lp-fix
./run_train.sh       # 14 training curves: d4, d4-ed, d4-ed-all, null, positive control, α = 0.95
B=../../../target/release/examples/ic_improve2_04_d4_2lp_fix
$B fit --train raw/train.jsonl --prior-train ../../ic_improve_20261014/01-d4-2lp/raw/train.jsonl --out raw/frozen.json
```

```
./run_test.sh        # 27 held-out curves
$B analyze --in raw/test.jsonl --prior ../../ic_improve_20261014/01-d4-2lp/raw/test.jsonl \
  --prior-summary ../../ic_improve_20261014/01-d4-2lp/raw/summary.json --out raw/summary.json
python3 -I plot.py
# from the repository root
D=research/ic_improve_20261015/04-d4-2lp-fix/ecbench
target/release/ecbench run --wait --allow-busy --cpus none --spec $D/spec-d4-ed.json --out $D/session-20261015-d4-ed
target/release/ecbench verify --dir $D/session-20261015-d4-ed --replay 8 --out $D/session-20261015-d4-ed.audit.json
target/release/ecbench compare --dir $D/session-20261015-d4-ed --a ic-k4-d4 --b ic-k4-d4-ed
```

Every job ran single-threaded under `nice`, `prlimit --as=3000000000` and `timeout 3600`. Conductor's control plane was unreachable (`conductor check` refused the connection), so there is no task card.

Files

- **New code:** `examples/ic_improve2_04_d4_2lp_fix.rs`, with the `train`, `fit`, `test` and `analyze` subcommands.
- **Edited, additively:**
  - `src/cryptanalysis/ecbench/fp4.rs`: the field `K4Params.eager_la`;
  - `src/cryptanalysis/ecbench/methods.rs`: the base `yline-d4-ed`.
- **Data:**
  - `raw/train.jsonl` (14 rows), `raw/frozen.json` and `raw/fit.stdout`;
  - `raw/test.jsonl` (27 rows), `raw/summary.json` and `raw/analyze.stdout`;
  - `ecbench/`.
- **Figures:** `figures/{paired_ratio,relations,phases}.{svg,png}`.
- **Report:** `REPORT.pdf`, built from this README.